A Counterexample to Formal Elementary-Symmetric Convergence in the Analytic Wiener Algebra

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Source question and answer

For the elementary symmetrization map

$$\pi : \mathbb{D}^d \longrightarrow \mathbb{C}^d, \qquad \pi(z) = (s_1(z), \dots, s_d(z)),$$

Section 7.1 of Sasane, arXiv:2201.00183, asks the following. Given $f \in W^+_{\mathrm{sym}}(\mathbb{D}^d)$, decompose $f = \sum_{k\geq 0} p_k$ into homogeneous symmetric polynomials and write each p_k as a polynomial in $s_1, \dots, s_d$. Does the resulting formal series $g \in \mathbb{C}[[s_1, \dots, s_d]]$ converge on a domain containing $\pi(\mathbb{D}^d)$, so that $f = g \circ \pi$?

Under the standard meaning of convergence of a multivariable power series, the answer is *no*. In fact, the illustrative function printed immediately after the question is already a counterexample. There is still a corrected positive result: retaining the transformed polynomials as weighted-homogeneous blocks always gives a locally uniformly convergent holomorphic factor on $\pi(\mathbb{D}^d)$.

Proof Intuition

The original Wiener norm sees the coefficient mass of a symmetric polynomial *before* changing coordinates. Rewriting in s_1, s_2 can create large coefficients which cancel inside each polynomial block. Along the diagonal $z = w = r$, the block values remain small, but taking absolute values of their individual monomials replaces cancellation by a binomial sum. That sum has exponential base $3r^2$, which exceeds one well inside the bidisc. A power series cannot converge on an open set through such a point.

Theorem 1 (full negative answer in dimension two)**.** *Let*

$$f(z, w) = \sum_{n=1}^{\infty} \frac{(z^2 + w^2)^n}{n^2 2^n}. \tag{1}$$

Then $f \in W^+_{\mathrm{sym}}(\mathbb{D}^2)$. Under $s_1 = z + w$, $s_2 = zw$, the formal series prescribed by the source is

$$g(s_1, s_2) = \sum_{n=1}^{\infty} \sum_{k=0}^{n} \frac{1}{n^2 2^n} \binom{n}{k} (-2)^k s_1^{2n-2k} s_2^k. \tag{2}$$

This series fails to converge absolutely at points of the symmetrized bidisc $G_2 := \pi(\mathbb{D}^2)$. Consequently it has no power-series convergence domain containing G_2.

Proof. The monomials belonging to different n in (1) have different total degrees. Hence there is no cross-block coefficient cancellation and

$$\|f\|_1 = \sum_{n=1}^{\infty} \frac{\|(z^2+w^2)^n\|_1}{n^2 2^n} = \sum_{n=1}^{\infty} \frac{2^n}{n^2 2^n} = \sum_{n=1}^{\infty} \frac{1}{n^2} < \infty.$$

The function is symmetric. The identity $z^2 + w^2 = s_1^2 - 2s_2$ and the binomial theorem give (2).

Fix $r \in (1/\sqrt{3}, 1)$ and take $(z,w) = (r,r)$. Its image $(s_1, s_2) = (2r, r^2)$ lies in G_2. The sum of the absolute values of all monomials in the nth block of (2), evaluated there, is

$$\frac{1}{n^2 2^n} \sum_{k=0}^{n} \binom{n}{k} 2^k (2r)^{2n-2k} (r^2)^k$$
$$= \frac{(4r^2 + 2r^2)^n}{n^2 2^n} = \frac{(3r^2)^n}{n^2}.$$

These quantities do not even tend to zero because $3r^2 > 1$. Also, a monomial with exponents $(2n-2k, k)$ has weighted degree $(2n-2k) + 2k = 2n$, so distinct pairs (n,k) give distinct monomials; the calculation does not overcount coefficients which later combine.

A multivariable power series is absolutely and locally uniformly convergent throughout the interior of its convergence set. Thus (2) cannot converge on any domain containing the open set G_2. $\square$

For the concrete choice $r = 3/4$, the absolute nth block is $(27/16)^n/n^2$.

What remains true: block factorization

The negative answer concerns the formal *monomial* series, not the existence of a holomorphic factor.

Proposition 1 (corrected positive statement)**.** *For every $d < \infty$ and every*

$$f(z) = \sum_{\alpha \in \mathbb{N}_0^d} c_\alpha z^\alpha \in W^+_{\mathrm{sym}}(\mathbb{D}^d),$$

let $p_k = \sum_{|\alpha|=k} c_\alpha z^\alpha$ and let Q_k be the unique polynomial satisfying $p_k = Q_k \circ \pi$. Then

$$F(s) := \sum_{k=0}^{\infty} Q_k(s)$$

converges locally uniformly on $G_d := \pi(\mathbb{D}^d)$, defines a holomorphic function there, and satisfies $f = F \circ \pi$.

Proof. The finite symmetrization map $\pi : \mathbb{D}^d \to G_d$ is proper. Given a compact $K \subset G_d$, its preimage is compact in $\mathbb{D}^d$, so there is $\rho < 1$ such that $|z_j| \le \rho$ for every $z \in \pi^{-1}(K)$ and every j. For $s = \pi(z) \in K$,

$$|Q_k(s)| = |p_k(z)| \le \rho^k \sum_{|\alpha|=k} |c_\alpha|.$$

The Weierstrass test proves uniform convergence on K. Every partial sum is a polynomial in s, so the limit is holomorphic. Substitution gives $F(\pi(z)) = \sum_k p_k(z) = f(z)$. $\square$

For Theorem 1, the block factor is

$$F(s_1, s_2) = \sum_{n=1}^{\infty} \frac{(s_1^2 - 2s_2)^n}{n^2 2^n} = \mathrm{Li}_2\left(\frac{s_1^2 - 2s_2}{2}\right).$$

It is holomorphic on G_2 because $(s_1^2 - 2s_2)/2 = (z^2 + w^2)/2$ has modulus less than one there. At the diagonal points used above, the polynomial blocks converge by cancellation, while their monomial absolute values diverge. This precisely separates Proposition 1 from the source's formal-series hope.

Novelty and verification

Exact-id, title, formal-series, elementary-symmetric-variable, and symmetrized-polydisc searches were run in the local FA/Banach indexes and primary arXiv sources through 2026-08-12. No later answer or duplicate packet was found. The coefficient calculation was independently checked with exact rational arithmetic; the source question and its example occur on PDF pages 17–18.

References

- A. Sasane, *Banach algebras of symmetric functions on the polydisc*, arXiv:2201.00183.